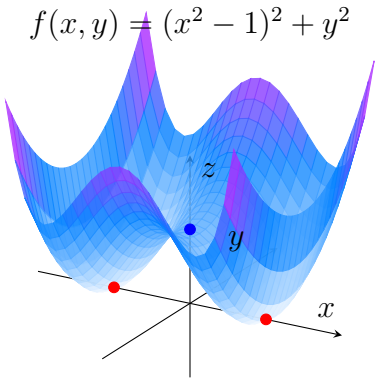
10 Optimization and Lagrange Multipliers

10.1 Extrema of Multivariable Functions

Definition 10.1: Local Maximum and Minimum

For a function $f(x, y)$ and a point (a, b) in its domain, we say that (a, b) is a

- **local maximum** if $f(a, b) \geq f(x, y),\ \forall (x, y)$ in a neighborhood of (a, b).
- **local minimum** if $f(a, b) \leq f(x, y),\ \forall (x, y)$ in a neighborhood of (a, b).



Definition 10.2: Critical Point

We call a point $P(a, b)$ a **critical point** of f if

$$\nabla f(P) = \vec{0}$$

Theorem 10.1: Critical Points and Local Extrema

If $P(a, b)$ is a local extremum of f, then P is a critical point of f.

For single variable functions, we can use the second derivative test to determine the nature of a point on the function. For $f(x)$, if $f'(a) = 0$, then:

- If $f''(a) > 0$, then f has a local minimum at $x = a$.
- If $f''(a) < 0$, then f has a local maximum at $x = a$.
- If $f''(a) = 0$, the test is inconclusive.

For multivariable functions, we can use its quadratic approximation to determine the nature

of a critical point.

Problem 10.1: Determining the Critical Points of a Function

Find critical points of the function $f(x, y) = x^4 + y^4 + 4xy$. Which are local maxima, local minima, and saddle points?

Solution 10.1

First, we find the critical points by setting the gradient of f to zero. The gradient is given by:

$$\nabla f(x, y) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle = (4x^3 + 4y, 4y^3 + 4x)$$

Setting each component to zero gives us the system of equations:

$$4x^3 + 4y = 0$$
$$4y^3 + 4x = 0$$

This implies $y = -x^3$. Substituting $y = -x^3$ into the second equation gives:

$$4(-x^3)^3 + 4x = 0 \implies -4x^9 + 4x = 0 \implies 4x(-x^8 + 1) = 0$$

This gives us $x = 0$ or $x^8 = 1$. The equation $x^8 = 1$ has solutions $x = \pm 1$. So the critical points are $(0, 0)$, $(1, -1)$, and $(-1, 1)$.

To determine the nature of these critical points, we write $f(x, y)$ in the quadratic approximation form around each critical point.

$$f(x, y) \approx a + bx + cy + Ax^2 + Bxy + Cy^2$$
$$\text{If } \nabla f(0, 0) = 0, \text{ then } b = c = 0, a = f(0, 0)$$
$$\implies f(x, y) \approx f(0, 0) + Ax^2 + Bxy + Cy^2$$

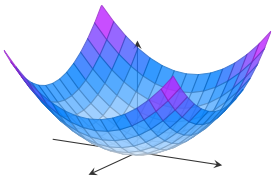
The second derivative is then given by

$$\frac{\partial^2 f}{\partial x \partial x} = 2A$$
$$\frac{\partial^2 f}{\partial y \partial y} = 2C$$
$$\frac{\partial^2 f}{\partial x \partial y} = B$$

Fact: If (a, b) is a critical point of f, then the quadratic approximation is given by

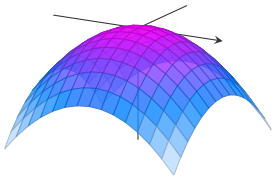
$$Q(x, y) = \frac{1}{2} f_{xx}(a, b)(x - a)^2 + f_{xy}(a, b)(x - a)(y - b) + \frac{1}{2} f_{yy}(a, b)(y - b)^2$$

There are three possible nondegenerate behaviors for a quadratic function near a critical point:



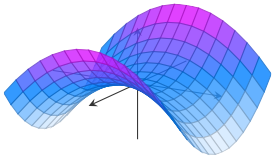
$Q(x, y) = x^2 + y^2$

Minimum



$Q(x, y) = -x^2 - y^2$

Maximum



$Q(x, y) = x^2 - y^2$

Saddle

For a general quadratic function

$$Q(x, y) = Ax^2 + Bxy + Cy^2,$$

its behavior is determined by $4AC - B^2$. For the quadratic approximation of f at a critical point (a, b),

$$A = \frac{1}{2} f_{xx}(a, b), \qquad B = f_{xy}(a, b), \qquad C = \frac{1}{2} f_{yy}(a, b).$$

Hence,

$$4AC - B^2 = f_{xx}(a, b) f_{yy}(a, b) - f_{xy}(a, b)^2.$$

Definition 10.3: Second Derivative Test

Let (a, b) be a critical point of f, and define

$$D = f_{xx}(a, b)f_{yy}(a, b) - f_{xy}(a, b)^2.$$

Then:

- If $D > 0$ and $f_{xx}(a, b) > 0$, then (a, b) is a **local minimum**.
- If $D > 0$ and $f_{xx}(a, b) < 0$, then (a, b) is a **local maximum**.
- If $D < 0$, then (a, b) is a **saddle point**.
- If $D = 0$, the test is **inconclusive**.

Example 10.1 (Usage of the Discriminant)**.**

We have $f(x, y) = x^2 + y^2 - 3xy$.

From calculation, we know that $\nabla f = \langle 2x - 3y, 2y - 3x\rangle$. Setting this equal to zero gives us the system of equations:

$$2x - 3y = 0 \quad \text{and} \quad 2y - 3x = 0.$$

Solving this system, we find that the only critical point is $(0, 0)$.

The discriminant is given by $D = (2)(2) - (-3)^2 = 4 - 9 = -5 < 0$.

We now know that $(0, 0)$ is a saddle.

Definition 10.4: Global Extrema

Given a function $f(x, y)$ defined on some domain $\mathcal{D}$, we say that $(a, b) \in \mathcal{D}$ is a

- **global maximum** if $f(a, b) \geq f(x, y),\ \forall (x, y) \in \mathcal{D}$.
- **global minimum** if $f(a, b) \leq f(x, y),\ \forall (x, y) \in \mathcal{D}$.

10.2 Domains and Boundaries

Definition 10.5: Interior and Boundary Points

Let $\mathcal{D} \subseteq \mathbb{R}^2$ be a domain.

- A point $(a, b) \in \mathcal{D}$ is an **interior point** if is has some disk around it in $\mathcal{D}$.
- A point $(a, b) \in \mathcal{D}$ is a **boundary point** such that every disk around it intersects $\mathcal{D}$ and $\mathbb{R}^2 \setminus \mathcal{D}$.

* $\mathbb{R}^2 \setminus D$ means the set of points in $\mathbb{R}^2$ that are not in D.

Definition 10.6: Closed and Bounded Sets

A set $\mathcal{D} \subseteq \mathbb{R}^2$ is **closed** if it contains all of its boundary points, and **bounded** if it can be contained in a disk of finite radius.

Theorem 10.2: Extreme Value Theorem

Let f be a continuous function on a closed and bounded set $\mathcal{D} \subseteq \mathbb{R}^2$.

Then f attains both a global maximum and a global minimum on $\mathcal{D}$.

Example 10.2 (Finding Global Extrema)**.**

Given the function $f(x, y) = x^2 + y^2 + xy$, and the domain $D = \{(x, y) : x^2 + y^2 \leq 4\}$.

We set the gradient of f to zero to find critical points:

$$\nabla f = \langle 2x + y, 2y + x \rangle = \langle 0, 0 \rangle .$$

This gives us the system of equations:

$$2x + y = 0 \quad \text{and} \quad 2y + x = 0.$$

Solving this system, we find that the only critical point is $(0, 0)$.

Since $\mathcal{D}$ is closed and bounded, and f is continuous on $\mathcal{D}$, the Extreme Value Theorem guarantees that f attains both a global maximum and a global minimum on $\mathcal{D}$. Therefore, we need to check the boundary of $\mathcal{D}$ to find the global extrema.

The boundary of $\mathcal{D}$ is the circle defined by $x^2 + y^2 = 4$. We can parameterize this boundary

using polar coordinates:

$$x = 2\cos(t), \quad y = 2\sin(t), \quad 0 \leq t < 2\pi.$$

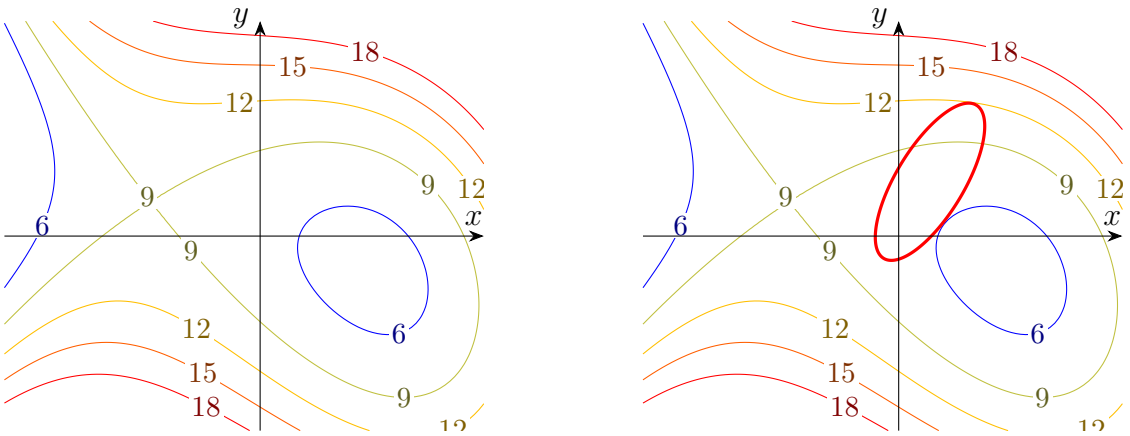
Hence,

$$\begin{aligned} f(2\cos(t), 2\sin(t)) &= (2\cos(t))^2 + (2\sin(t))^2 + (2\cos(t))(2\sin(t)) \\ &= 4\cos^2(t) + 4\sin^2(t) + 4\cos(t)\sin(t) \\ &= 4 + 4\cos(t)\sin(t) \\ &= 4 + 2\sin(2t). \end{aligned}$$

Therefore, the maximum value of f on the boundary is $4 + 2 = 6$ at $x = 2, y = 0$, and the minimum value is $4 - 2 = 2$ at $x = -2, y = 0$.

10.3 Lagrange Multipliers

Example 10.3 (Extrema of a Function on a Curve)**.**
Below is a function f and a curve C given by $g(x, y) = c$.



In this example, the extrema under constraints occur where the level curves of f are tangent to the constraint curve C. This is a particularly lucky situation, as we can visually identify the points of tangency.

Theorem 10.3: Lagrange Multipliers

Let $f(x, y)$ be a function and $g(x, y) = c$ be a constraint curve. If $g(x, y) = c$ is tangent to a level curve of f, then

$$\nabla f \parallel \nabla g \implies \nabla f = \lambda \nabla g$$

Therefore, to find the extrema of f under the constraint $g(x, y) = c$, we can solve the system of equations:

$$\nabla f = \lambda \nabla g, \quad g(x, y) = c$$

for x, y, and λ, assuming that $\nabla g \neq 0$ at the points of interest.

Example 10.4 (Using Lagrange Multipliers)**.**

We want to find the extrema of $f(x, y) = x + 2y$ subject to the constraint $3x^2 + 2y^2 = 7$.

First, we compute the gradients:

$$\nabla f = \langle 1, 2 \rangle, \quad \nabla g = \langle 6x, 4y \rangle.$$

Setting $\nabla f = \lambda \nabla g$ gives us the system of equations:

$$\begin{cases} 1 = \lambda(6x) \\ 2 = \lambda(4y) \\ 3x^2 + 2y^2 = 7 \end{cases}$$

Solving the first two equations for λ, we get:

$$\begin{cases} \lambda = \dfrac{1}{6x} \\ \lambda = \dfrac{1}{2y} \end{cases}$$

Setting these equal gives us:

$$\frac{1}{6x} = \frac{1}{2y} \implies 2y = 6x \implies y = 3x.$$

Substituting $y = 3x$ into the constraint equation gives:

$$\begin{aligned} 3x^2 + 2(3x)^2 &= 7 \\ 3x^2 + 18x^2 &= 7 \\ 21x^2 &= 7 \\ x^2 &= \frac{1}{3} \\ x &= \pm\frac{1}{\sqrt{3}}. \end{aligned}$$

Thus, the corresponding y values are:

$$y = 3\left(\pm\frac{1}{\sqrt{3}}\right) = \pm\sqrt{3}.$$

Therefore, the points of interest are:

$$\left(\frac{1}{\sqrt{3}}, \sqrt{3}\right) \text{ and } \left(-\frac{1}{\sqrt{3}}, -\sqrt{3}\right).$$

Example 10.5 (Another Example of Lagrange Multipliers)**.**
Given a function $Q(K, L) = AK^{\alpha}L^{1-\alpha}$. This describes the output when K units of capital and L units of labor are used, where $A > 0$ is a constant and $0 < \alpha < 1$.

Suppose that $A = 1$ and $\alpha = \frac{3}{5}$. Given a budget of \$12 with the cost of capital being \$3 per unit and the cost of labor being \$4 per unit, we want to maximize $Q(K, L)$ subject to the constraint $2K + 3L = 12$.

The process is similar to the previous example. We first rewrite the constraint as $g(K, L) = 2K + 3L - 12 = 0$. Then, we compute the gradients:

$$\begin{aligned} \nabla Q &= \left\langle \frac{3}{5}K^{-2/5}L^{2/5}, \frac{2}{5}K^{3/5}L^{-3/5} \right\rangle \\ \nabla g &= \langle 2, 3 \rangle. \end{aligned}$$

Setting $\nabla Q = \lambda \nabla g$ gives us the system of equations:

$$\begin{cases} \frac{3}{5} K^{-2/5} L^{2/5} = 2\lambda \\ \frac{2}{5} K^{3/5} L^{-3/5} = 3\lambda \\ 2K + 3L = 12 \end{cases}$$

Solving the first two equations for λ, we get:

$$\frac{3}{10} K^{-2/5} L^{2/5} = \frac{2}{15} K^{3/5} L^{-3/5} \implies 9L = 4K \implies L = \frac{4}{9} K.$$

Substituting $L = \frac{4}{9}K$ into the constraint equation gives:

$$\begin{aligned} 2K + 3\left(\frac{4}{9}K\right) &= 12 \\ 2K + \frac{4}{3}K &= 12 \\ \frac{10}{3}K &= 12 \\ K &= \frac{36}{10} = 3.6. \end{aligned}$$

Then, substituting $K = 3.6$ into $L = \frac{4}{9}K$ gives:

$$L = \frac{4}{9} \cdot 3.6 = 1.6.$$

Therefore, the optimal allocation of resources is $K = 3.6$ units of capital and $L = 1.6$ units of labor.